

Complex Diabetes Case Study 1: Rural Farmer with Multiple Complications

Case Overview

Patient Profile: 58-year-old male farmer, Kumar (pseudonym), living in a rural village near Tumkur.

Presenting Complaint: Worsening fatigue, frequent urination, and non-healing foot ulcer for 3 months.

History of Present Illness

Presenting Symptoms:

- Polyuria (frequent urination) for past 6 months
- Polydipsia (excessive thirst)
- Fatigue and weakness interfering with farming work
- Blurred vision for 2 months
- Weight loss of 8 kg in 4 months despite good appetite
- Non-healing ulcer on right big toe for 3 months

Past Medical History:

- Diagnosed with diabetes 12 years ago during routine health camp
- Irregular treatment - takes metformin when "feels sick"
- No formal diabetes education received
- No regular blood glucose monitoring
- Family history: Mother died of "sugar disease" complications

Social History:

- 45-year-old farmer, owns 2 acres of land
- Daily physical labor: 8-10 hours farming work
- Family: Wife (housewife), 2 children (15 and 12 years old)
- Tobacco: Chew tobacco 5-6 packets/day for past 20 years
- Alcohol: Occasional (festive drinking)
- Education: Primary school education only
- Monthly income: ₹8,000-10,000

Current Medications:

- Metformin 500mg - takes irregularly (2-3 times/week)
 - No other medications
 - Uses traditional remedies occasionally
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Physical Examination

Vital Signs:

- BP: 160/95 mmHg (multiple readings)
- Pulse: 84/min regular
- Temperature: 98.4°F
- BMI: 22.1 kg/m² (underweight)
- Waist circumference: 88 cm

General:

- Pale conjunctiva, decreased skin turgor
- Evidence of nutritional deficiency
- Mild dehydration

Systemic Examination:

Cardiovascular: Normal heart sounds, no murmurs **Respiratory:** Normal vesicular breath sounds **Abdomen:** Soft, non-tender, no organomegaly

Neurological:

- Peripheral neuropathy: Vibration sense reduced in both feet
- Loss of ankle reflexes bilaterally
- Pain sensation present

Foot Examination:

- Right big toe: 2x2 cm ulcer with necrotic base
- Surrounding cellulitis with erythema
- No odor or discharge
- Pedal pulses palpable
- Monofilament testing: Reduced sensation

Vision:

- Best corrected visual acuity: 6/24 right eye, 6/18 left eye
- Cataract changes evident

Laboratory Investigations

Blood Glucose:

- Random blood glucose: 280 mg/dL
- HbA1c: 11.5% (very poor control)
- Fructosamine: 385 µmol/L (elevated)

Lipid Profile:

- Total cholesterol: 220 mg/dL
- LDL: 140 mg/dL
- HDL: 32 mg/dL (low)

- Triglycerides: 220 mg/dL

Kidney Function:

- Serum creatinine: 1.4 mg/dL
- eGFR: 52 mL/min/1.73m² (Stage 3 CKD)
- Urine albumin: 150 mg/g creatinine (macroalbuminuria)
- Urine ketones: Negative

Other Tests:

- ECG: Normal sinus rhythm, no ischemic changes
- Hb: 10.8 g/dL (anemia of chronic disease)
- Vitamin B12: 180 pg/mL (borderline low)

Diagnosis

Primary Diagnosis:

- Type 2 Diabetes Mellitus with poor glycemic control (HbA1c 11.5%)
- Multiple chronic complications:
 - Diabetic peripheral neuropathy
 - Diabetic nephropathy (Stage 3 CKD)
 - Diabetic foot ulcer with cellulitis
 - Suspected diabetic retinopathy

Associated Conditions:

- Hypertensive cardiovascular disease
- Dyslipidemia
- Tobacco-related oral health issues
- Mixed nutritional deficiencies
- Depressive symptoms (undiagnosed)

Management Plan

Immediate Interventions (Community Level):**1. Wound Care & Infection Control:**

- Daily dressing changes with saline and povidone-iodine
- Oral antibiotics (amoxicillin-clavulanate for 10 days)
- Elevation and rest instructions
- Daily wound monitoring

2. Glycemic Control:

- Regular insulin initiation (basal-bolus regimen education)
- SMBG teaching (3 times daily)

- Diet counseling for nutritional rehabilitation
- Family involvement in diabetes management

3. **Blood Pressure Management:**

- Antihypertensive addition (ACE inhibitor preferred)
- Lifestyle modifications emphasis
- Regular monitoring protocol

Referral Strategy:

1. **Primary Care Level:** Initial management and stabilization
2. **Secondary Care Level:** Diabetes specialist consultation within 1 week
3. **Tertiary Care Level:** Nephrology referral for CKD management
4. **Specialist Referrals:**
 - Ophthalmologist for retinopathy screening
 - Vascular surgeon if wound deteriorates
 - Nutritionist for comprehensive dietary planning

CBME Learning Objectives

Patient Care Competencies:

- Perform systematic multisystem examination in diabetic patients
- Identify and prioritize multiple complications
- Develop coordinated referral strategy
- Provide culturally appropriate health education

Medical Knowledge:

- Understand natural history of diabetes complications
- Apply complication screening protocols
- Recognize red flags requiring urgent referral

Communication Skills:

- Explain complex medical conditions to low-literacy patients
- Coordinate multidisciplinary care team
- Educate family members effectively

Systems-Based Practice:

- Navigate limited resources for maximum patient benefit
- Coordinate across primary/secondary/tertiary care levels
- Access available government diabetes schemes

Discussion Points for Students:

1. **Priority Setting:** How would you prioritize multiple complications in resource-limited settings?
 2. **Cultural Context:** What cultural factors might affect treatment adherence in this case?
 3. **Multidisciplinary Care:** Which team members should be involved and why?
 4. **Preventive Lens:** How could earlier intervention have prevented these complications?
 5. **Resource Management:** What government programs can help this patient?
 6. **Family Dynamics:** How will Kumar's farming occupation impact diabetes management?
 7. **Health System Navigation:** What are the challenges in referral systems in Indian context?
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References for Further Reading:

- IDF Position Statement on Management of Type 2 Diabetes
- NCDIR Guidelines for Management of Diabetes in India
- WHO PEN (Package of Essential Noncommunicable Disease Interventions)
- RSSDI Guidelines for Diabetes Management

Faculty Guide: Use this case to discuss real-world challenges in rural diabetes care and develop student competencies in comprehensive diabetes management.